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Unit 3 - Designing Solutions

PDF Documents

PDF Document 1: 1738153459698-Chapter 9 - Unit 3 - Designing Solutions.pdf

This PDF document is included in the unit materials.

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UNIT 3 DESIGNING SOLUTIONS

Programming Concepts

WHILE LOOP

A "while" loop continues to execute its block of code **as long as the condition is true**. It checks the condition before running the loop's code. If the condition is true, the loop runs. If it's false, the loop stops. For example, "while the sun is up, keep the LED off."

✓ WHILE



condition

While the sun is up,



task

Keep the LED off

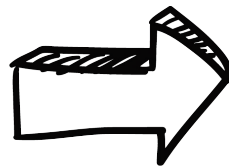
Another Example:

"Think of using a 'while' loop like brushing your teeth while a timer is running. You start brushing, and 'while' the timer hasn't beeped (meaning time is not up yet), you keep brushing. As soon as the timer beeps, signaling the time is up, you stop brushing."



condition

While the timer is running,



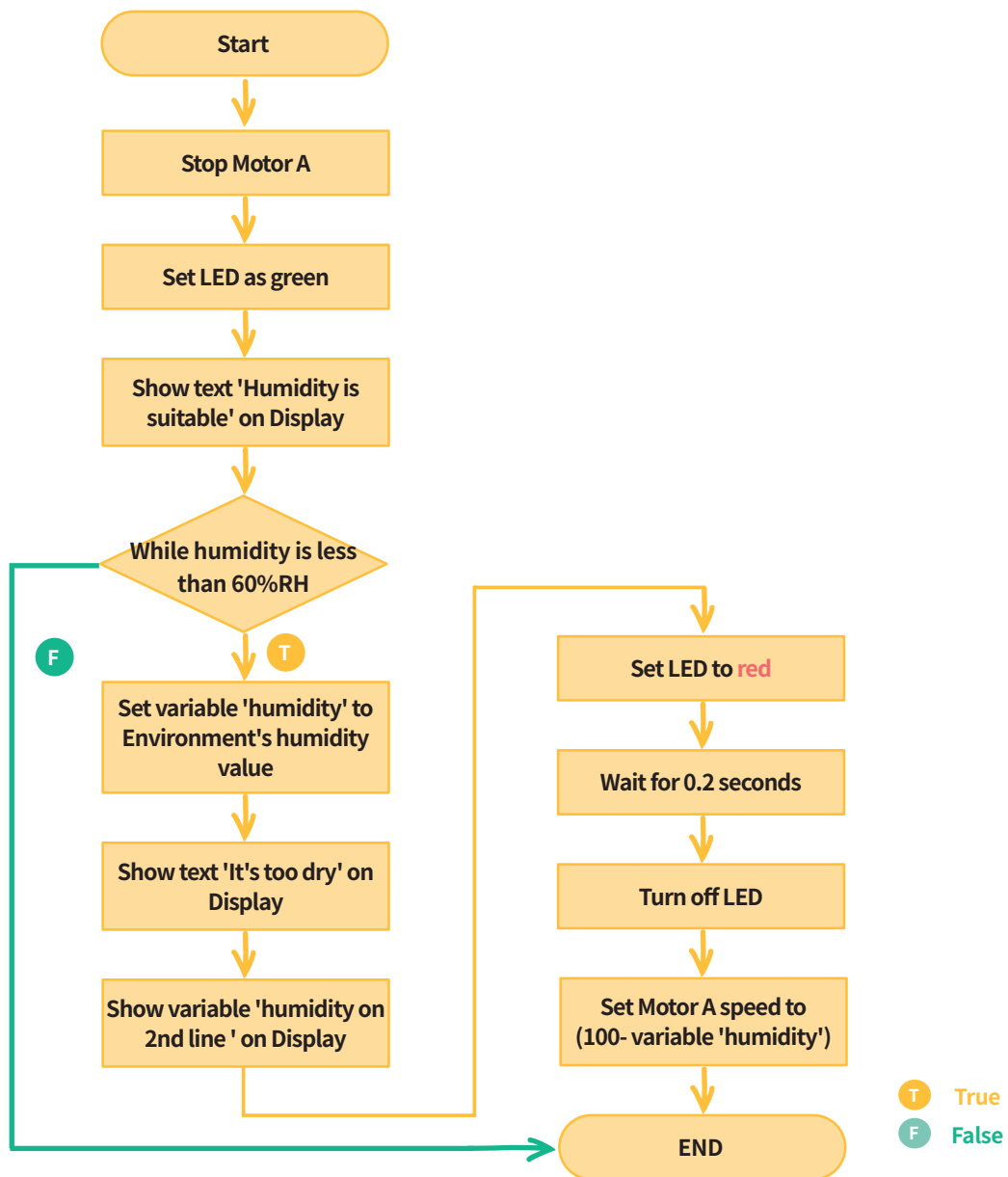
task

Keep brushing your teeth.

Flowcharts & Algorithms

Let's read the flowchart and create the code.

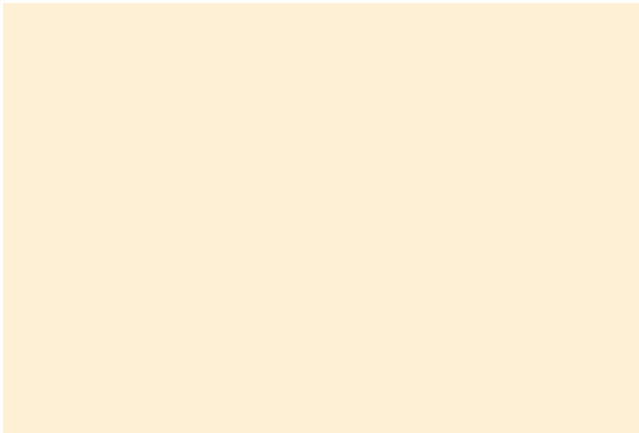
FLOWCHART



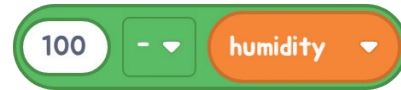


DEBUGGING

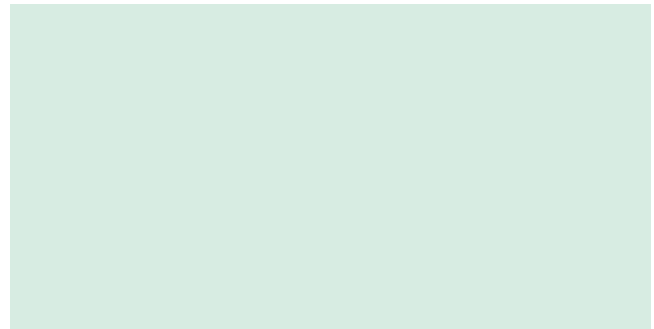
1 Which blocks is initialization codes in here?



2 What will happen if we change the code



'-' to '+'?

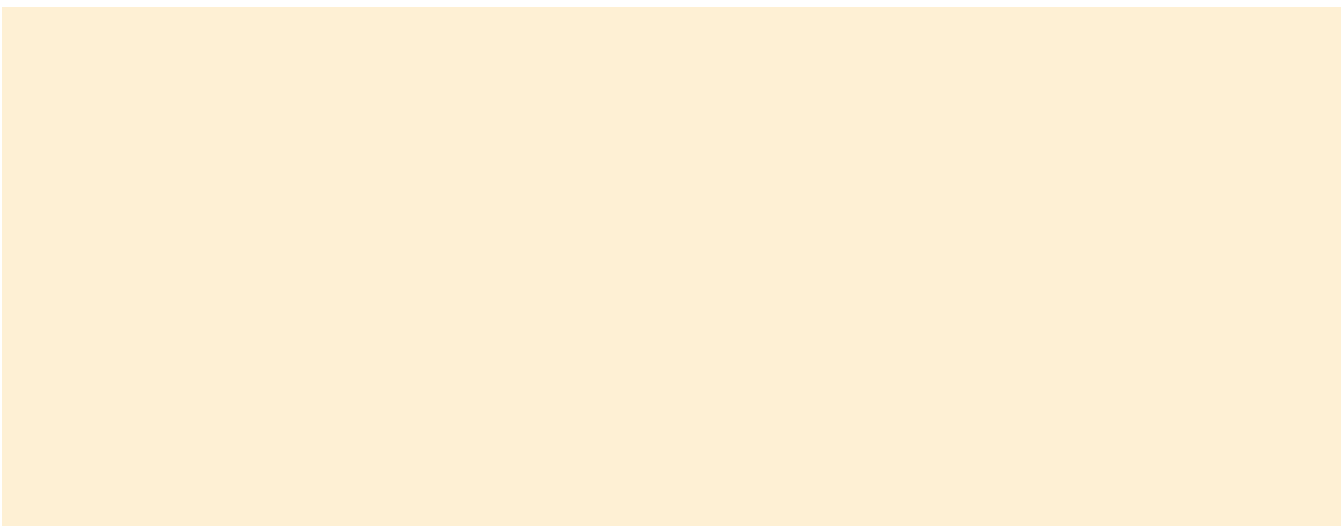


MISSION 1

1 What is the role of this block?

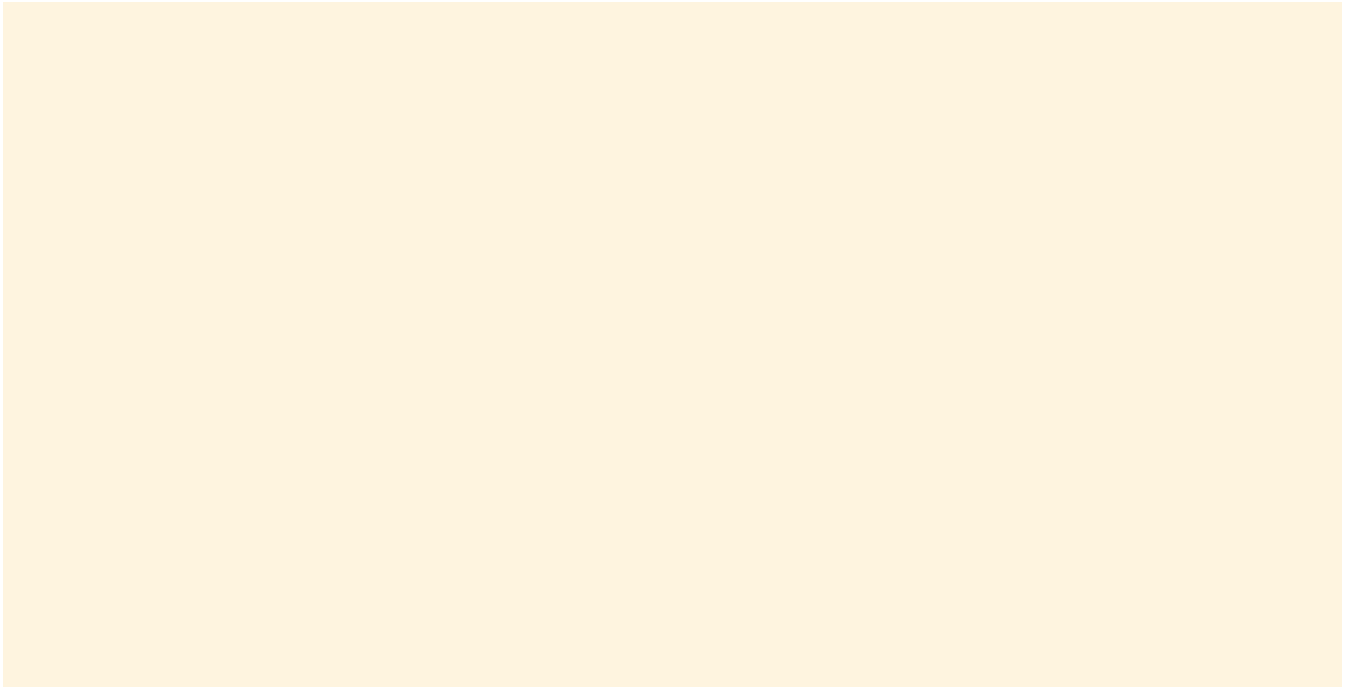


2 What is the role of this block?



MISSION 2

- 1 Add speakers to make a sound when the motor is moving.



- 2 When the Button is on, watering system has to work unless it has to stop. Use and operator to add Button block.

